

Metal-free boron-doped carbon microspheres as excellent cathode catalyst for rechargeable Li–O₂ battery

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Abstract Rechargeable Li–O₂ batteries are attracting more and more interest due to their high energy density. Meanwhile, the replacement of high-cost and scarce precious-metal catalysts has attracted more and more attention. Currently, many academic researchers have paid attention to find highly efficient metal-free catalysts as air cathode material. Herein, the boron-doped carbon microspheres (B-CMs) were prepared through a novel and facile static calcination method and showed high electrocatalytic activity as a cathode material. The battery with a B-CM cathode delivered a high initial discharge capacity of 13,757.2 mAh g^{−1} and outstanding coulombic efficiency of 90.1 % at 100 mA g^{−1}. In addition, stable cyclability (151 cycles with stable discharge voltage of ~2.60 V with a cutoff capacity of 1000 mAh g^{−1} at 200 mA g^{−1}) has been exhibited. These performances are due to three main points: boron carbide compound changed the surface area of the CMs and formed the mesopore architectures as well as the large surface area of 683.738 m² g^{−1}; the reduce of boron atom can slow down the oxidation of the CMs during the cyclings; and finally, the electron-deficient boron atom introduction greatly facilitated

Li⁺ diffusion and electrolyte immersion and enhanced the oxygen reduction and evolution reaction kinetics.

Keywords Boron-doped carbon microspheres · Air cathode · Li–O₂ battery

Introduction

Rechargeable Li–O₂ battery has attracted extensive attentions due to its superior theoretical specific energy of nearly 11,680 Wh kg^{−1}, which equals gasoline and has 5–10 times greater energy density than that of lithium-ion battery [1–4]. A typical rechargeable non-aqueous Li–O₂ battery is comprised of a Li metal anode, a non-aqueous electrolyte, and a porous air cathode. Various works indicated that the battery performance strongly depended on the air cathode [5]. The discharge product (Li₂O₂) stored in the air cathode can block the electrolyte and oxygen pathways, which also limits the rate capacity, coulombic efficiency, and cycle life of the battery. In order to harvest high performance Li–O₂ batteries, it is urgently needed to design a porous air cathode material with good activity on the oxygen reduction reaction (ORR) and the oxygen evolution reaction (OER) [7, 35]. Recently, it is found that large surface area and mesopore volume of the cathode material tend to improve the capacity and cyclability significantly [6]. Mesopores can effectively improve the electrolyte immersion and provide more active sites for electrochemical reactions, while large surface area can facilitate a Li / O₂ reaction, which are critical for reversible conversion as well as the oxygen reduction and evolution reaction [14]. Therefore, it is critical to develop a cathode with a good catalytic surface and microstructure to enhance the reaction kinetics and hold a maximum amount of reaction products.

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Many works have been made to design porous cathode materials. Previous studies have reported that conventional precious metal (Au, Pt, Pd, etc.)-based materials are the most active catalysts for high capacity and low overpotential [8–10]. However, the high cost, poor stability, and high toxicity prevent their widespread use. In the energy shortage twenty-first century, replacement of the high-cost and scarce precious-metal catalysts with commercially available alternatives has attracted much attention [11–13]. What is more, the battery is easy to achieve stable cyclability without any metal particles fading during cycling. Thus, recently, much interest has focused on developing a rational metal-free catalyst material.

Current interesting metal-free catalyst is sp^2 -hybridized carbon and its derivatives. Different kinds of sp^2 -hybridized carbon were designed as air cathodes, including carbon nanotubes (CNTs), carbon microspheres (CMs), carbon foams, graphene, and carbon nanofibers (CNBs) [1, 15–18]. Among those various structural design types, carbon microspheres are preponderant for their high surface area as well as macropores and mesopore presence. Wu et al. reported that graphene-based hollow spheres electrochemically catalyzed the oxygen reduction effectively [19]. And we have reported that the pure CMs can even act as a cathode material and achieved a good activity in our previous work [24]. In addition to various porous structure designs, it is also reported that the surface chemistry of carbon materials also performs important functions in Li–O₂ batteries. Both computations and detailed experiments proved that doping foreign atoms on those traditional carbon materials is an effective way to achieve better performances [6]. Heteroatom-doped carbon is recently emerging as a promising lithium-air battery cathode material for its ability to tailor the chemical and electron nature of carbon-based materials [20]. Huge efforts have demonstrated that nitrogen-doped carbon materials (such as carbon nanotubes or graphene) as a Li–O₂ battery cathode material showed higher capacity and low overpotential [13, 20, 21]. Other heteroatoms, such as boron, sulfur, and phosphorus, are also of interest, more specifically to the electron-deficient boron dopant for the introduction of additional p-type carriers in carbon systems [22]. It has proved that the substitutional boron in carbon materials could boost the ORR and OER activities because electron-deficiency boron atoms are easy to attract electrons from carbon atoms, break the C–C bond, and join carbon lattice at substitutional site of grapheme layers to rearrange the surface of the carbon material. It is reported that boron-doped multi-walled carbon nanotubes have exhibited high performances for both ORR and OER in alkaline solutions, with better catalytic activity than Pt [29]. It is also found that higher percent of boron concentration showed more improved catalytic performance [23].

However, the available cathode catalysts have not promoted high specific capacity, high efficiency, and good cycle stability of a Li–O₂ battery to a satisfactory level.

Here, we prepared metal-free boron-doped carbon microspheres (B-CMs) by a facile and novel static calcination method. After boron doping, the composites showed a mesopore structure with big surface area of 683.738 m² g^{−1}. Li–O₂ batteries with B-CM material cathodes showed predominant performance with an available initial discharge capacity of 13,757.2 mAh g^{−1} and a cutoff capacity of 3000 mAh g^{−1} for 15 cycles under a 100 mA g^{−1} current density. Moreover, the B-CM battery delivered excellent cycling stability and good performance after 151 cycles with a low overpotential under a 200 mA g^{−1} current density. Therefore, it should be claimed that metal-free B-CMs had great potential to be a promising cathode material for Li–O₂ battery.

Experimental

Catalyst preparation

The B-CMs were synthesized by the following steps: (1) 5 wt% sucrose dissolved in 6 M H₂SO₄ solution was heated to reflux for 10 h, and then the mixture was washed with deionized water to obtain the CM precursor; (2) the CM precursor and the boric acid were mixed with a mass ratio of 1:15 and dissolved by deionized water, followed by a 15-min ultrasonic blending process; (3) the mixture solution was then stirred at room temperature for 10 h and evaporated, and the composite powder was subjected to thermal annealing under nitrogen at 900 °C for 2 h in a muffle furnace and cooled to room temperature in nitrogen atmosphere; and (4) the above samples were activated at 400 °C for 1 h in air to obtain the B-CMs. For comparison, the CMs were also synthesized without adding boron acid using the same method.

Material characterization

The as-prepared sample was physically characterized using a transmission electron microscope (TEM, JEM100CXII), field emission scanning electron microscope (FE-SEM, S-4800), energy dispersive X-ray (EDX) spectrometer attached to the FE-SEM instrument, X-ray diffraction (XRD, D/MAX-2500), Raman spectroscopy (in Via Reflex), Fourier transform infrared spectra (FTIR, FTS3000), and Brunauer–Emmett–Teller (BET) specific surface (NOVA2200e, USA).

Preparation of electrode and Li–O₂ cell assembling configuration

The cathode was prepared by brushing a mixture of the prepared B-CMs, AB (acetylene black) and polyvinylidene

fluoride (PVDF) in *N*-methyl-2-pyrrolidone with a mass ratio of 70:10:20 wt% onto a carbon paper (purchased from Shanghai Hesen, 0.3 mm thickness). The assembling configuration and test system of the cells were constructed by the method previously reported [24].

Results and discussion

Structure of the B-CMs

Figure 1 presents the as-prepared B-CM morphology and microstructure. TEM and SEM images show the as-prepared material in the presence and absence of the boron dopant. Figure 1a, c shows that the diameters of the spheres became larger and more uniform in size (500 nm) after boron doping, due to attachment by the boron. The rough surface (Fig. 1b)

stems from the etching of boron compared with CMs' smooth edge (Fig. 1d). The SEM images show that many nanosized holes appeared on the spheres surface (Fig. 1e) after boron doping compared with the pure CMs (Fig. 1f), because of some boron oxide (mainly B_2O_3) evaporation during the calcination procedure in high temperature. The holes effectively improved the electrolyte immersion and provided more active sites for electrochemical reactions [25, 26]. The composite was further characterized by N_2 adsorption–desorption experiments (Fig. 2a). The specific surface area of the B-CMs reached $683.738 \text{ m}^2 \text{ g}^{-1}$, higher than that of CMs ($626.919 \text{ m}^2 \text{ g}^{-1}$) in Fig. 2c. The pore size distribution is centered at 3–9 nm after boron doping (Fig. 2b) compared with 2–5 nm of the CMs (Fig. 2d). The larger surface area and mesopore structure not only facilitated the electrolyte immersion and Li^+ diffusion but also provided a large volume critical for O_2 diffusion and O_2/Li_2O_2 conversion [25–28].

Fig. 1 TEM images of B-CMs (a, b) and CMs (c, d). SEM images of the B-CMs (e) and CMs (f)

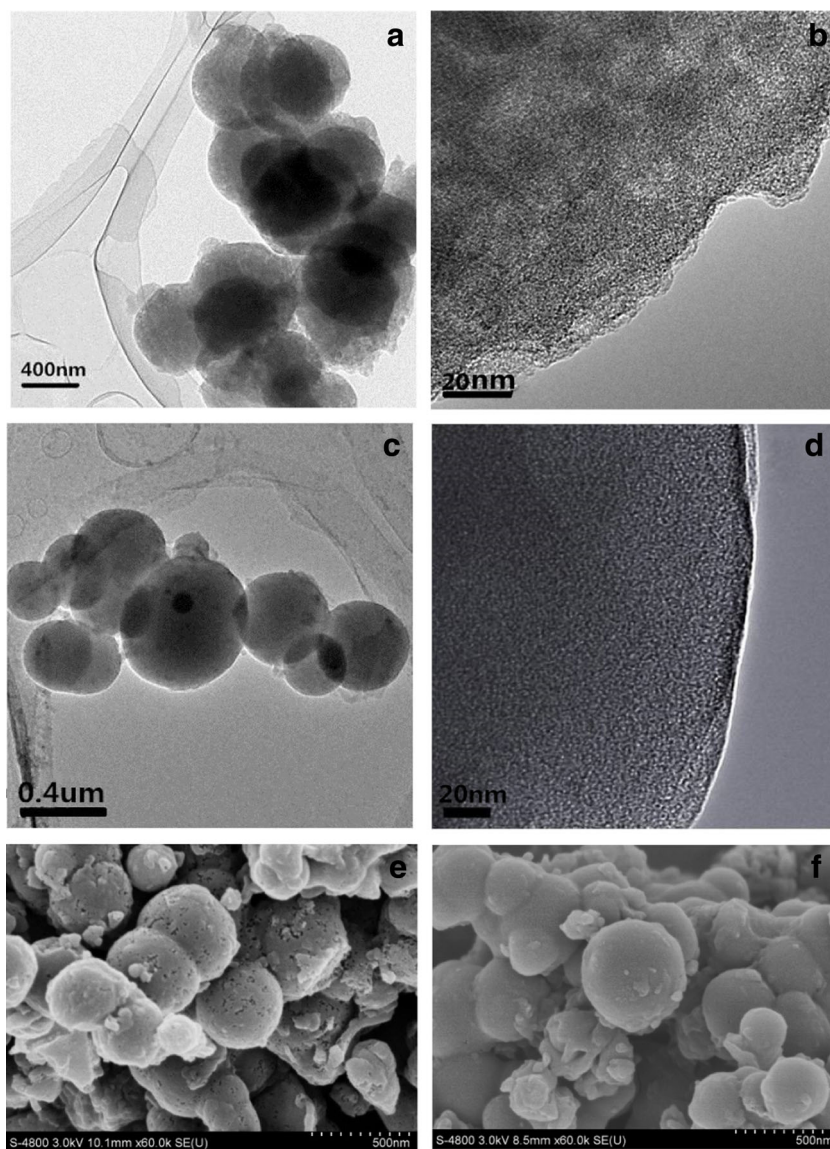
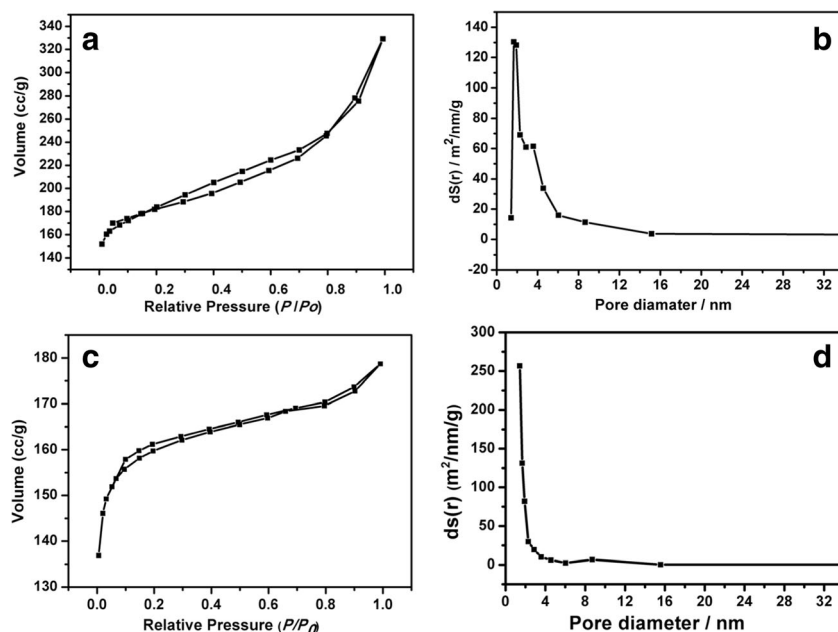


Fig. 2 N₂ adsorption–desorption isotherms of B-CMs (a, b) and CM (c, d)



Actually, B-CMs synthesized by the static calcination method contained rich B elements decomposed during the calcinating process. EDX confirmed the existence of boron (8.07 %) in Fig. S1. The boron content amount is much higher than other boron-doped carbon materials made by the thermal annealing method [29] and chemical vapor deposition (CVD) floating catalyst method [23].

The typical FTIR spectra (Fig. S2) also suggest that B was successfully coupled to C and O. The absorption at 1450 cm^{-1} is attributed to B–O stretching vibration and the strongly enhanced absorption at 1190 cm^{-1} compared with that of CMs is the B–C absorption band. The bonding of B–O–C is supported by the absorption peak at 1027.17 cm^{-1} . The results indicate that the existence of B is mostly B_3C and partly BC_2O [30].

Figure 3a shows the XRD patterns of the as-prepared B-CM and CM samples. The two obvious diffraction peaks centered at $2\theta = 23.3^\circ$ and 43.9° of all samples correspond to the (002) and (100) carbon phase reflections, demonstrating that B-CMs have the same crystal structure as the CMs. However, the diffraction peaks of the B-CMs are broadened and the intensities of these peaks become weakened compared with

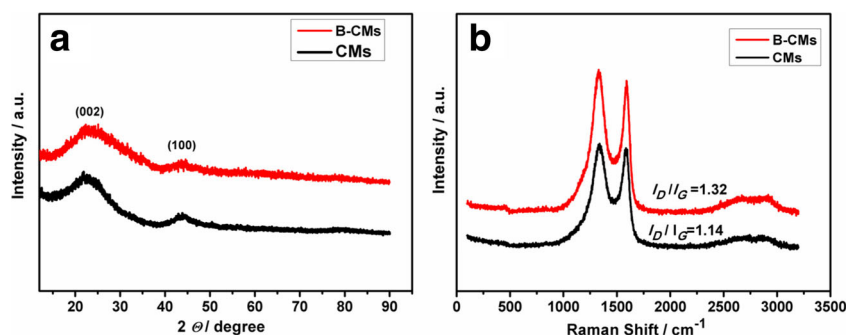
the CM peaks, suggesting that the carbon becomes more disordered after B doping.

The Raman spectroscopy further studied the structural defects induced by B doping in the carbon spheres. Figure 3b shows that all the samples exhibit two strong peaks at about 1586 cm^{-1} (G-band) and 1343 cm^{-1} (D-band), corresponding to the graphitic layers and the defects in the graphitic layers, respectively. Figure 3b also gives the ratios of D-band to G-band in intensity (I_D/I_G) for the two samples. The I_D/I_G is 1.32 for B-CMs and 1.14 for CMs, indicating more defects are introduced in carbon microspheres after boron doping. This is in good agreement with the XRD results.

Electrochemical characteristics of the Li–O₂ cells

B-CMs present many outstanding structural and compositional characters and demonstrate possible Li–O₂ battery applications. The discharge and charge measurements were performed in the voltage range of 2.0–4.5 V (vs. Li/Li⁺). Both the applied current (mA g^{-1}) and achieved capacity (mAh g^{-1}) were normalized to the weight of B-CMs and the CMs, respectively. The catalytic activity of B-CMs toward ORR and

Fig. 3 XRD patterns of the B-CMs and CMs (a) and Raman spectra of the B-CMs and CMs (b)



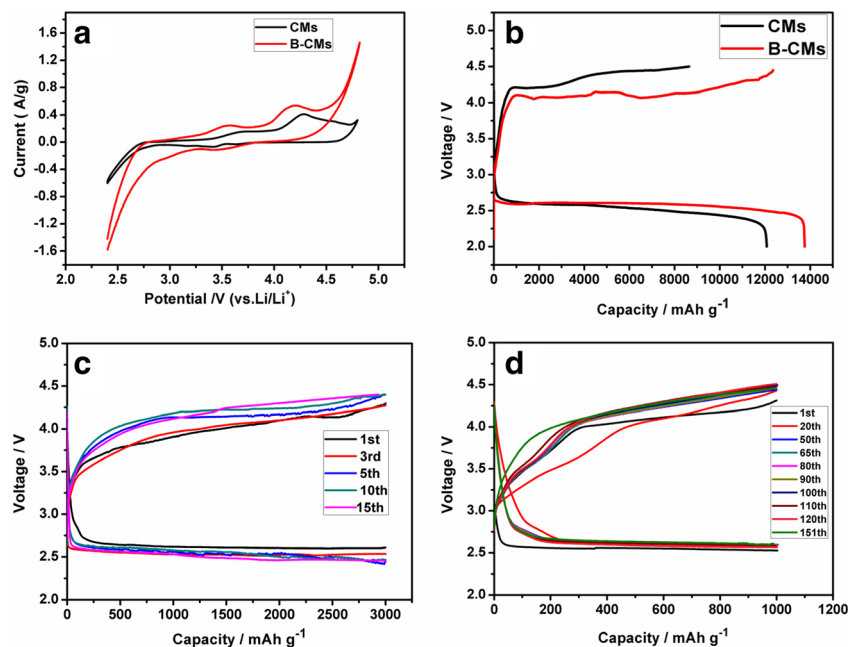
OER was explored by cyclic voltammetry in Li–O₂ batteries, under O₂ atmosphere, at a scan rate of 0.1 mV s^{−1}. The CM cathode was tested as a comparison. Figure 4a shows a significantly higher current density for ORR is obtained for the B-CM electrode, indicating a higher discharge capacity in the presence of boron. In addition, the B-CM cathode also displays a higher current density for OER during the anodic scan demonstrating the high reversibility of the cell. It is worth noting that B-CM cathode exhibited lower ORR and higher OER onset potential as well as bigger peak area compared with the individual CM cathode. Those results reveal that the B-CM catalyst possessed superior catalytic activity and enhanced the ORR and OER processes. Figure 4b shows the first discharge–charge curves of the Li–O₂ batteries with a current density of 100 mA g^{−1}. The Li–O₂ batteries based on the CM cathode can yield an initial discharge capacity of 12,081 mAh g^{−1}. While the B-CM cathode battery delivered a capacity of 13,757.2 mAh g^{−1} with a lower charge potential (4.05 V) and a higher discharge potential (2.62 V). The coulombic efficiency of the B-CMs was 90.1 % which is higher than Co and N co-doped graphene [31]. We also tested the battery based on B-CMs with a cutoff capacity of 3000 mAh g^{−1}, at 100 mA g^{−1}, as Fig. 4c shows. The first cycle showed a charge potential of 4.05 V and a discharge potential of 2.62 V. The battery maintained this high capacity and stable discharge platform (2.60 V) for 15 cycles, which has not been reported for Li–O₂ batteries with single heteroatom-doped carbon materials as the cathode material.

As for the cyclic tests, the cells were discharged and charged with a cutoff capacity of 1000 mAh g^{−1} at a current density of 200 mA g^{−1}. The cells showed good performance over 151 cycles and the Fig. 4d depicts the corresponding

profiles. Obviously, B-CMs showed stable low charge voltage at 4.08 V in the first cycle. What is more, in the 151th cycle, the discharge voltages (2.60 V) almost stayed the same as the first cycle, and the charge profile is almost unchanged after the 50th cycle. The CM cathode battery just sustained 60 cycles with a higher overpotential and unstable platform only after 30 cycles by the same test method in our previous report [24]. The superior performance of the B-CM cathode is attributed to the additional p-type carriers in CM systems by boron doping and the presence of a larger surface area and mesopore structure, which provided optimized oxygen transport network for O₂ diffusion and O₂/Li₂O₂ conversion. Also, the boron sites in the structure of B-CMs modified the electronic structure of boron-doped carbon and act as the O₂ chemisorption sites [32].

The air cathode SEM images after discharge and charge during the first cycle were also taken to investigate the B-CM composite performance. The characterizations of charge–discharge process were all limited to the cutoff capacity of 1000 mAh g^{−1} at 200 mA g^{−1}. After the first discharge (Fig. S3a), the insoluble species precipitated on the surface of B-CMs, and the Li₂O₂ was also observed with the typical nanowall shaped in previous reports [33]. After the subsequent charge process, a relatively clear spherical structure on the cathode was recovered in the B-CM cathode (Fig. S3b). This was further proved by XRD patterns in Fig. S3c. The peaks of Li₂O₂ could be clearly observed in the discharge electrode. After the subsequent charge process, the Li₂O₂ characteristic peaks nearly disappeared, indicating good battery reversibility. Beside the XRD patterns, the Raman spectroscopy is also shown in Fig. S3d. The peak of Li₂O₂ is observed at 849.48 cm^{−1}. The weak peak at 1135.67 cm^{−1} is

Fig. 4 CV curve of Li–O₂ batteries with CMs and B-CMs at a scan rate of 0.1 mV s^{−1} (a). The initial full discharge–charge curves of the CM and B-CM electrodes at a current density of 100 mA g^{−1} (b). Curtailing capacity of 3000 mA g^{−1} with B-CM electrode at a current density of 100 mA g^{−1} (c). Cycling curves with B-CM electrode at a current density of 200 mA g^{−1} (d)



assigned to $\text{CH}_3\text{CO}_2\text{Li}$ which is reported commonly with ether-based electrolytes when carbon materials are used as the air cathode [34]. In addition, the I_D/I_G ration of B-CM cathode is much higher in the first recharge than in the first discharge, which indirectly confirms that B-induced defects in the CMs recovered after the charging process, and the discharged products can be efficiently decomposed during the charging process. These results verify the superior B-CM activity in electrochemical cyclic stability during discharge–charge processes.

Conclusion

In summary, we have prepared metal-free B-CMs with a high content of electron-deficient boron (8.07 %) through a facile static calcination method as efficient catalyst for non-aqueous Li-O_2 batteries. The Li-O_2 batteries with a B-CM cathode material are capable of high capacity and super round trip efficiency as well as stable cyclability. The excellent performance is due to the enhanced surface and architecture of carbon microspheres by the introduction of the electron-deficient boron atoms. B-CMs with large surface area of $683.738 \text{ m}^2 \text{ g}^{-1}$ and mesopore structure have enhanced the kinetics of ORR and OER in Li-O_2 batteries. Therefore, metal-free B-CMs are promising low-cost and high-efficiency Li-O_2 battery cathode materials and provide a new outlook for the application of the materials to Li-O_2 cells.

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